Universitat
Pompeu Fabra
Barcelona

Lecture 8. Channel capacity

8.1 Introduction

So far we have seen examples of channel codes, such as the repetition code or the (7,4) Hamming code, which aim at transmitting information reliably over a communication channel using the data transmission block diagram depicted in Figure 8.1 below. With such system, we transmit $k = \log_2 M$ information bits using n channel uses, where n is the length of the sequences $\boldsymbol{x}_m$ and $\boldsymbol{y}$.

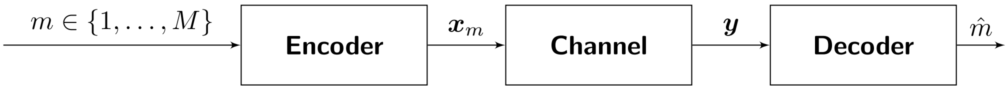

Figure 8.1: Encoding and decoding functions in channel coding.

We wish to design codes able to transmit information at a rate $R = \frac{k}{n}$ as high as possible with an error probability P_{e} the smallest as possible. We say that reliable communication is possible at a rate R if $\lim_{n\to\infty} P_{\mathrm{e}} = 0$ at a fixed rate R. The largest rate R at which reliable communication is possible is called the **channel capacity**.

8.2 Channel capacity

The capacity of a discrete memoryless channel with input $X \in \mathcal{X}$, output $Y \in \mathcal{Y}$ and transition probability $P_{Y|X}(y|x)$, denoted by C and derived by Shannon in 1948, is given by

$$C = \max_{P_X(x)} I(X;Y), \tag{8.1}$$

where the maximization is over all possible input distributions, and $I(X;Y)$ is the **mutual information** between the input X and the output Y, given by

$$I(X;Y) = \sum_{x\in\mathcal{X}} \sum_{y\in\mathcal{Y}} P_X(x) P_{Y|X}(y|x) \log_2 \frac{P_{Y|X}(y|x)}{P_Y(y)}. \tag{8.2}$$

In other words, to find the capacity of a discrete memoryless channel, one needs to calculate the mutual information in (8.2) for all possible input distributions $P_X(x)$ and find the input distribution that maximizes the mutual information. The resulting maximum value of the mutual information is the capacity of the channel C, and the input distribution that leads to the channel capacity is called the capacity-achieving input distribution, denoted as $P_X^\star(x)$. The mutual information (8.2) can be interpreted as the amount of information that a random variable Y has about another random variable X, or it can also be understood as the reduction in information once we observe Y as we will see in the next section.

There are two important implications with respect to the mutual information and reliable communication:

(a) For a fixed input distribution $P_X(x)$, reliable communication is possible as far as $R < I(X;Y)$ because there exist codes such that $P_{\mathrm{e}} \to 0$ as $n \to \infty$.

(b) Conversely, for $R > I(X;Y)$, reliable communication is not possible as there is no code such that $P_{\mathrm{e}} \to 0$ as $n \to \infty$.

Before seeing detailed example on the calculation of the channel capacity of a binary symmetric channel, we establish a number of relations between the mutual information and entropies.

8.3 Mutual information and entropies

The mutual information in (8.2) can be interpreted as the expected value of the random variable $Z = f(X,Y)$, the function of two variables $f(x,y) = \log_2 \frac{P_{Y|X}(y|x)}{P_Y(y)}$. That is, $I(X;Y) = \mathbb{E}[f(X,Y)]$. Using that the joint distribution of X and Y can be written as

$$P_{XY}(x,y) = P_{Y|X}(y|x)P_X(x) = P_{X|Y}(x|y)P_Y(y), \tag{8.3}$$

and expanding the logarithm, we obtain that the mutual information can be written in various ways, including

$$\begin{aligned} I(X;Y) &= \mathbb{E}\left[\log_2 \frac{P_{Y|X}(y|x)}{P_Y(y)}\right] = \mathbb{E}\big[\log_2 P_{Y|X}(y|x)\big] - \mathbb{E}\big[\log_2 P_Y(y)\big] && (8.4)\\ &= \mathbb{E}\left[\log_2 \frac{P_{XY}(x,y)}{P_X(x)P_Y(y)}\right] = \mathbb{E}\big[\log_2 P_{XY}(x,y)\big] - \mathbb{E}\big[\log_2 P_X(x)\big] - \mathbb{E}\big[\log_2 P_Y(y)\big] && (8.5)\\ &= \mathbb{E}\left[\log_2 \frac{P_{X|Y}(x|y)}{P_X(x)}\right] = \mathbb{E}\big[\log_2 P_{X|Y}(x|y)\big] - \mathbb{E}\big[\log_2 P_X(x)\big] && (8.6)\\ &= I(Y;X), && (8.7) \end{aligned}$$

implying that the mutual information is a symmetric quantity since $I(X;Y) = I(Y;X)$. It is easy to observe that many of the terms in the expressions above are expectations of the logarithm of probability distribution, which are indeed **entropies**! Let us define the following quantities

$$H(X) = -\sum_{x\in\mathcal{X}} P_X(x)\log_2 P_X(x) \tag{8.8}$$

$$H(X,Y) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P_{XY}(x,y)\log_2 P_{XY}(x,y) \tag{8.9}$$

$$H(Y|X) = -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P_{XY}(x,y)\log_2 P_{Y|X}(y|x) \tag{8.10}$$

where $H(X)$ is the entropy of X, $H(X,Y)$ is the joint entropy of X,Y and $H(Y|X)$ is the conditional entropy of Y given X, which can be rewritten as

$$\begin{aligned} H(Y|X) &= -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} P_X(x)P_{Y|X}(y|x)\log_2 P_{Y|X}(y|x) && (8.11)\\ &= \sum_{x\in\mathcal{X}} P_X(x)\Big(-\sum_{y\in\mathcal{Y}} P_{Y|X}(y|x)\log_2 P_{Y|X}(y|x)\Big) && (8.12)\\ &= \sum_{x\in\mathcal{X}} P_X(x)H(Y|X=x) && (8.13) \end{aligned}$$

where

$$H(Y|X=x) = -\sum_{y\in\mathcal{Y}} P_{Y|X}(y|x)\log_2 P_{Y|X}(y|x) \tag{8.14}$$

is the conditional entropy of Y for a specific x. Entropy, beyond being the fundamental limit of data compression, has an interpretation of the uncertainty present in a given random variable. Using the definitions above (watch out the signs!), we can write equataions (8.4), (8.5), (8.6) and (8.7) as

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) && (8.15)\\ &= H(X) + H(Y) - H(X,Y) && (8.16)\\ &= H(X) - H(X|Y) && (8.17)\\ &= I(X;Y). && (8.18) \end{aligned}$$

We end this section by stating three important properties of the entropy and the mutual information:

1. The mutual information is a non-negative quantity, that is $I(X;Y) \geq 0$ with equality if and only if X and Y are independent.
2. The entropy is also a non-negative quantity that is upper-bounded as $H(X) \leq \log_2 |\mathcal{X}|$, with equality if and only if X has equiprobable symbols.
3. Conditioning reduces entropy, that is $H(Y|X) \leq H(Y)$, suggesting that knowing another random variable X can only reduce the uncertainty present in Y.

8.4 Binary symmetric channel

We will now calculate the capacity of the binary symmetric channel. Since the input is binary, we let $X \in \{0, 1\}$ with probability distribution $\{q, 1-q\}$, that is, $P_X(0) = q$ and $P_X(1) = 1 - q$. Since $P_X(x)$ only depends on one parameter q, that satisfies $0 \le q \le 1$, we can rewrite the capacity expression in (8.1) as

$$C = \max_{0\le q\le 1} I(X;Y). \tag{8.19}$$

In order to compute (8.19), we start from the convenient expression of the mutual information in (8.15), that is $I(X;Y) = H(Y) - H(Y|X)$. For the binary symmetric channel with bit flipping probability p we have $P_{Y|X}(0|1) = P_{Y|X}(1|0) = p$. Therefore, the conditional entropy $H(Y|X)$ can be calculated as

$$\begin{aligned} H(Y|X) &= -\sum_{x\in\{0,1\}}\sum_{y\in\{0,1\}} P_X(x)P_{Y|X}(y|x)\log_2 P_{Y|X}(y|x) && (8.20)\\ &= \underbrace{q\Big[(1-p)\log_2\frac{1}{1-p} + p\log_2\frac{1}{p}\Big]}_{x=0} + \underbrace{(1-q)\Big[(1-p)\log_2\frac{1}{1-p} + p\log_2\frac{1}{p}\Big]}_{x=1} && (8.21)\\ &= \underbrace{(1-p)\log_2\frac{1}{1-p} + p\log_2\frac{1}{p}}_{h_2(p)}. && (8.22) \end{aligned}$$

The last expression is the binary entropy function, that we denoted as $h_2(p)$.

In addition, we know that the entropy of the output $H(Y)$ satisfies $H(Y) \le \log_2(2) = 1$ and that $H(Y) = 1$ is achieved with equality when Y has equal probability. By the total probability law, the distribution of the output is $P_Y(0) = q(1-p) + (1-q)p$. If the input distribution is $P_X(0) = q = \frac{1}{2}$, then $P_Y(0) = \frac{1}{2}$ and therefore $H(Y) = 1$ achieves the maximum value as possible independently on p. Since $H(Y|X)$ in (8.22) does not depend on q, we conclude that $P_X^\star(0) = \frac{1}{2}$ is the capacity-achieving input distribution and that the channel capacity is

$$C(p) = 1 - h_2(p) \text{ bits/channel use.} \tag{8.23}$$

We illustrate (8.23) in Figure 8.2.

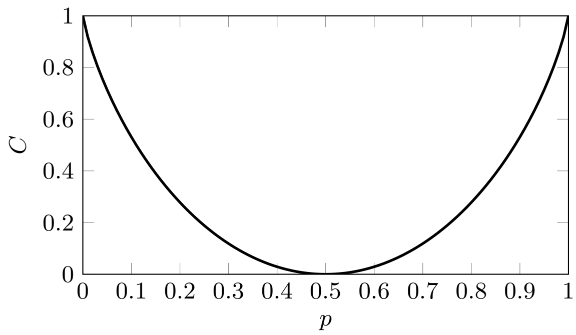

Figure 8.2: Channel capacity of the binary symmetric channel.

There are three final interesting points to notice:

- If $p = 0$, there are no bit flips, the channel is noiseless and the capacity $C = 1$ bit/channel use reliably. In this case, there is no need to code.
- If $p = \frac{1}{2}$, the channel flips bits with equal probability and the best one can do at the decoder is to guess.
- Observe the symmetry for $p > \frac{1}{2}$. For example, for $p = 1$ where all bits are flipped, it suffices to revert the output of the channel deterministically and we can still reliably transmit 1 bit/channel use.
- If $p = 0.11$ we have that $C = 0.5$, i.e., if we wish to reliably transmit with a code of rate $R = \frac{1}{2}$ we will only be able to do so if $p < 0.11$.